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Differentiating Parkinson Disease Subtypes Using Clinical Balance Measures

Morgane Prime, BSc[1,3], **J. Lucas McKay, PhD, MSCR**[2], **Allison Bay, MPH**[3], **Ariel Hart, BS**[3], **Chaejin Kim, PharmD**[3], **Amit Abraham, PhD, BPT**[3,4], **Madeleine E. Hackney, PhD.**[3,5,6]

[1]Faculty of Biology, University of Toulouse III - Paul Sabatier, Toulouse, France

[2]Department of Biomedical Engineering, Emory University and the Georgia Institute of Technology, Atlanta, Georgia, USA

[3]Division of General Medicine and Geriatrics, Department of Medicine, Emory School of Medicine, Atlanta, Georgia, USA

[4]Department of Kinesiology, University of Georgia, Athens, GA, USA

[5]Atlanta VA Center for Visual and Neurocognitive Rehabilitation, Decatur, Georgia, USA

[6]Department of Rehabilitation Medicine, Emory School of Medicine, Atlanta, Georgia, USA

Abstract

BACKGROUND AND PURPOSE: People with Parkinson disease (PD) present phenotypes that are characterized as tremor-dominant (TD) or postural instability/gait difficulty (PIGD) subtypes. Differentiation of subtypes allows clinicians to predict disease course and adjust treatment. We examined whether brief mobility and balance measures can discriminate PIGD from TD phenotypes.

METHODS: We performed a cross-sectional study with individuals with PD (N=104). Blinded raters assessed participants with the Unified Parkinson Disease Rating Scale (UPDRS) or Movement Disorders Society revision (MDS-UPDRS), and balance assessments: 360° turn test, one-leg stance, a reactive postural control test and tandem walk. Participants were classified as PIGD or TD based on the UPDRS or MDS-UPDRS assessment results. Differences in balance variables between subtypes were assessed with univariate analyses. Receiver operating characteristic (ROC) curve analyses were performed to investigate the ability of balance variables to differentiate PD subtypes.

RESULTS: No differences between subtypes were observed for tandem walk or reactive postural control. PIGD participants performed worse on number of steps and time to complete the 360° turn test and on one-leg stance time. ROC curves showed only the 360° turn test discriminated PIGD from TD with high specificity (0.84). Post-hoc analyses revealed that the 360° turn test is the most discriminatory for classifying PD subtypes in early stages. ROC analyses based on

Corresponding author: Madeleine E. Hackney, phone: (404)321-6111, x 5006, mehackn@emory.edu, madeleine.hackney@va.gov, fax:(404)728-6425.

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combined models including both the 360° test and tandem walk test performance increased specificity to 0.97.

DISCUSSION AND CONCLUSIONS: The 360° turn test requires minimal time to administer and may be useful in mild-moderate PD for distinguishing PIGD from TD subtypes. **Video Abstract available** for more insights from the authors (See Video, Supplemental Digital Content 1).

Keywords

Parkinson disease; motor subtypes; postural instability and gait difficulty; tremor dominant; 360° turn test

INTRODUCTION

People with Parkinson Disease (PD) experience bradykinesia, rigidity, resting tremor, and postural instability. Such motor impairments lead to functional disabilities, including high risk of falls. [1] Variability in PD symptoms have resulted in the observation of two distinct PD phenotypes referred to as postural instability and gait difficulty (PIGD) and tremor-dominant (TD) subtypes. [2] Each has distinct motor features and prognoses. [3] PIGD phenotype is characterized by bradykinesia and rigidity in movement [4] whereas persons with tremor-dominant (TD) phenotype typically display resting tremor, normal gait, and mild disease progression. [5] People with PIGD experience significantly greater subjective intellectual, motor and occupational impairment than people with TD [2] and have a higher incidence of dementia [6] and neuropsychiatric disorders such as depression [7] and apathy [8]. More rapid disease progression is more common in the PIGD subtype[2]

Algorithms developed to distinguish TD and PIGD phenotypes use motor exam (Part III) scores from The Unified Parkinson's Disease Rating Scale (UPDRS) or the revised Movement Disorder Society- UPDRS (MDS-UPDRS).[2, 9, 10] Differentiation between TD and PIGD parkinsonism allows clinicians to tailor interventional strategies that accommodate the person's unique symptomatology and changing needs.[11, 12] Although the responsiveness to both subthalamic nucleus (STN) and globus pallidus internus (GPi) deep brain stimulation (DBS) is similar among the two motor subtypes, persons with TD may respond better to GPi DBS with respect to gait and persons with PIGD display less overall benefit from stimulation.[13] The ratio of the mean UPDRS tremor scores to the mean UPDRS PIGD scores are used to define persons with TD (ratio 1.5) and persons with PIGD (ratio 1.0), and scores between these cutoffs are referred to as "indeterminate." Although a reliable way to classify persons by motor subtype,[9] the UPDRS or the MDS-UPDRS-III take approximately twenty minutes (at minimum) to administer, and are not routinely administered in physical therapy practice

Balance tests have been used to predict PD-related complications, including fall risk. [14] The Fullerton Advanced Balance (FAB) scale, Mini-Balance Evaluation Systems Test (Mini-BESTest) and Berg Balance Scale (BBS) moderately distinguish fallers from non-fallers in individuals with PD. Specific items of these balance scales may predict future falls, especially using a recommended model combining "tandem stance," "one-leg stance," "rise

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to toes," "compensatory stepping backward," "turning 360°" and "placing foot on stool" when analyzing postural control deficits. [14] Studies comparing postural control in non-fallers versus fallers have shown that persons with PD who fall have reduced reactive postural control [15-17] and impaired ability to perform tandem stance/walk. [18] Fallers have poorer ability to stand on one leg,[19] and increased number of steps and time taken to complete a 360° turn may be a compensatory strategy used to avoid balance loss and prevent falls. [20]

Functional measures are advantageous because they can be quickly administered when a comprehensive neurological examination is infeasible or when item-level motor examination results are unavailable. Additionally, balance and postural stability tests are routinely used by physical therapists, making them a more viable option to assess clinical symptoms of PD. Using these brief tests to identify a persons' PD phenotypes with high sensitivity and specificity would enable clinicians to more efficiently determine intervention plans.

Functional tests have not been evaluated for ability to distinguish between TD and PIGD phenotypes. Therefore, our first aim was to compare performances of individuals with TD and PIG subtype on physical tests derived from common balance measures. The second aim was to examine whether these brief physical examinations could differentiate TD and PIGD phenotypes. As PIGD subtype is associated with more functional disability than TD subtype, [2] persons with PIGD were expected to have worse performance, i.e. taking more time and steps to complete a 360° turn, standing less time on one leg, execute tandem walk with more interruptions, and requiring more steps to recover balance during reactive postural control. We expected that performance on functional tests, e.g., number of steps required to complete a 360° turn, would be able to differentiate PIGD from TD subtypes in logistic regression models. Finally, we performed logistic regressions post-hoc to evaluate whether simple combinations of balance variables could better differentiate participants with PIGD from others in early stages of PD, i.e., Hoehn & Yahr Stage ≤ II.

METHODS

Study design

This study was approved by the Institutional Review Board at Emory University School of Medicine and the Research & Development Committee of the Atlanta Veterans Affairs (VA). Participants provided written informed consent before participating. This cross-sectional study was conducted using baseline measures of participants enrolled in a longitudinal trial.

Participants

Participants were recruited through the VA registry, the VA Informatics and Computing Infrastructure database, the FoxFinder website, the Movement Disorders unit of Emory University, PD newsletters, support groups and educational events and through word of mouth. Participants had a clinical diagnosis of PD made by a movement disorders specialist based on the United Kingdom PD Society Brain Bank diagnostic criteria. [21] Participants exhibited three of four PD cardinal signs (postural instability, tremor, rigidity and bradykinesia), had unilateral onset of symptoms, and had shown symptomatic benefit from

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antiparkinsonian medications, e.g., levodopa.[22] Participants were aged 40 years and older, were stage I-IV in the Hoehn & Yahr scale, and could walk three meters or more with or without an assistive device.

Clinical and Demographic Variables

Each participant was assessed with a standardized test battery including measurements of disease severity, global cognition[23, 24] and physical function (Composite Physical Function Index, "CPF").[25] Testing was performed in the "OFF" state (i.e., at least 12 hours after their last antiparkinsonian medication) to reduce the impact of medication-related motor fluctuations.

Classification of PD Phenotype

The motor exam component (Part III) of either the UPDRS[26] or the MDS-UPDRS[10] was administered and used to classify each participant as PIGD or TD. The data collection period occurred concomitant with the initial release and subsequent validation of the Movement Disorders Society (MDS) revised version of the UPDRS-III motor scale.[10] To accommodate changes in MDS recommended scales, participants enrolled earlier in data collection were assessed with the UPDRS-III. Participants enrolled later in data collection were assessed with the MDS-UPDRS-III to improve the clarity of the UPDRS instructions and to capture several non-motor aspects of PD. Total UPDRS-III scores were transformed into MDS-UPDRS-III scores before univariate analyses using published methods.[10]

Using UPDRS or MDS-UPDRS scores we classified participants as either PIGD or TD, per Stebbins et al.[9] Among those participants assessed with the MDS-UPDRS (n=74), a tremor subscore was assembled as the mean value of items II.10, III.14 LUE, III.15 RUE, III.16 LUE and RUE. A PIGD subscore was assembled as the mean value of left upper extremity (LUE), right upper extremity (RUE), left lower extremity (LLE), and right lower extremity (RLE) items II.12, II.13, III.10, III.11 and III.12. Among participants assessed with the UPDRS (n=30), the tremor subscore was assembled as the mean value of items II.16, III.20 face, RUE, LUE, RLE and LLE, III.21 RUE and LUE, and the PIGD subscore was assembled as the mean value of items II.13, II.14, II.15, III.29 and III.30. The ratio of the tremor to PIGD subscores was then used to classify participants as TD (>1.15), PIGD (<0.90), or otherwise indeterminate.

Balance variables used to determine PD phenotype

Balance variables used to determine PD phenotype were derived from four items of the FAB assessment as follows: The 360° turn test, tandem walk, one-leg stance, and reactive postural control.

- 360° turn test: Two variables, maximum time to turn and maximum steps to turn were derived from this test. Participants were asked to turn both right and left 360°, with a pause in between turns. The investigator the number of steps, and measured time needed for each turn with a stopwatch. Guarding for balance perturbations was provided by another investigator.

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- Tandem walk: The number of interruptions occurring during 10 steps of tandem walking was counted. Participants walked 10 feet forward along a pre-measured, highly visible line of duct tape, placing one foot directly in front of the other such that the heel and toe were in contact on each step forward. Interruptions occur when the participant (a) takes one or more steps away from the line or (b) is unable to achieve correct heel-to-toe position during any step.
- One-leg stance: The maximum time participants could balance on one leg was measured. Participants were asked to fold their arms across their chest, lift a leg off the floor without touching the other leg and stand with eyes open for 30 seconds.
- Reactive postural control: The maximum steps required to recover balance were counted during the push and release test. Participants slowly leaned back into rater's hands until the rater removed their hands.

Data analysis

All available data were used for participants, including those for whom data for some assessments are unavailable. Data were inspected for normality with statistical software (SPSS version 24, IBM Corp., Armonk, NY) The 360° turn test steps, tandem walk, one-leg stance, reactive postural control, and MoCA all had skewness and kurtosis statistics less than the widely acceptable +/−2 consistent with a normal distribution.[27] The skewness and kurtosis statistics for the 360° turn test time did not follow a normal distribution due to outliers who needed more than ten seconds to complete the test (n=9). With these data removed, skewness and kurtosis were 1.00 and 0.56, respectively. However, as long turn times were largely due to freezing of gait (FOG), a common symptom of PD and the PIGD phenotype, the outliers were ultimately included in analyses because outlier exclusion would limit external validity. Twelve participants classified as 'indeterminate' were excluded from further analyses to detect clear differences between 'TD' and 'PIGD' groups.

Separate univariate logistic regression analyses were applied to identify associations between PD phenotype (PIGD vs. TD, with TD as the reference group) and each of the candidate variables: Number of Steps to Turn 360°, Time to Turn 360° (s), Number of Interruptions in Tandem Walk, Maximal Time Standing on One Leg (s), Number of Steps to recover balance during Reactive Postural Control. Logistic regression models were used to create receiver operating characteristic (ROC) curves for each balance variable which plotted the sensitivity vs. 1-specificity of the model to identify PIGD participants.[28]

Identification of optimal cutoff values

Two optimal cutoff values (OCs) were identified for each balance variable. These OCs maximized different numerical combinations of sensitivity and specificity in identifying PIGD participants. As the cutoff value varied from the lowest value to the highest value of the balance variable in the dataset, sensitivity values decreased and specificity values increased.[29] We selected the OC value based on the resulting sensitivity and specificity according to two criteria. In Criterion 1, the cutoff value was the value of the balance variable that maximized the "Youden Index," which weights sensitivity and specificity

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equally.[30] The cutoff value was thus the value that maximized the sum (1.0 × sensitivity) + (1.0 × specificity) among all observed values of the balance variable. The Criterion 1 approach results in the fewest classification errors. However, minimizing errors may not always be appropriate if identifying as many cases as possible is the objective.[31] Therefore, in Criterion 2, the cutoff value was the value of the balance variable that maximized a second index that weighted sensitivity 50% more heavily than specificity to increase the ability to identify PIGD cases. In this case, we selected the cutoff value as the value that maximized the sum (1.5•sensitivity) + (1.0•specificity). These two cutoff values are labeled "Criterion 1" and "Criterion 2," respectively in the tables. All logistic regression results were reviewed to ensure that extreme values of balance variables did not influence classification and resulting sensitivity/specificity.

Post-hoc analyses

Post-hoc analyses examined 1) whether simple combinations of tests could improve overall sensitivity and specificity of the resulting classifier, and 2) the ability of the most discriminative test – the 360° turn test – to correctly differentiate between PD subtypes among those in Hoehn & Yahr Stages ≤ II. Because aggressive, early therapies may help people with PD with PIGD-type disease, given an arguably more severe prognosis, it is important to identify PIGD cases early in the disease. These analyses used the UPDRS as the gold standard.

Statistical tests were performed at alpha=0.05. RStudio software (version 1.0.153) was used for analyses, which were cross-verified in SAS software version 9.4.

RESULTS

One hundred four persons with PD with the TD or PIGD phenotype were included. Clinical and demographic features are summarized in Table 1 (See also Table S1, Supplemental Digital Content 2,).

Males represented 58% of the sample, which is representative of PD epidemiology.[32] Mean age and disease duration were 68±9 and 7±5 years, respectively. Median disease severity was Hoehn & Yahr Stage II (range, I-IV). PIGD participants reported more freezing episodes per week,more falls in the previous six months and were more likely to use assistive devices for walking than TD participants. TD participants had better composite physical function scores, MoCA scores and self-reported quality of life.

Univariate associations between PD phenotype and physical measures are provided in Figure 1 (see Table S2, Supplemental Digital Content 3).

PIGD participants needed more time (p=0.002) and more steps (p<0.001) to complete the 360° turn test and had shorter one-leg stance times (p=0.006).

ROC curves are presented in Figure 2.

Statistically-significant associations between balance variables and PD phenotype were identified for number of steps to complete the 360° turn test (p=0.001, AUC=0.751), time to

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complete the 360° test (p=0.007, AUC=0.740), and one-leg stance time (p=0.017, AUC=0.635). Only the 360° turn test had moderate or better accuracy to discriminate PD phenotype, as determined by AUC >0.70, with the other tests considered low (AUC ≤ 0.70).[33] Cutoff values suggested based on Criteria 1 and 2 are presented in Table 2.

A cutoff value of >7 steps during the 360° turn test could identify participants with PIGD with good sensitivity (0.831). A cutoff value of >10 steps could identify participants with PIGD with reduced sensitivity (0.523) but good specificity (0.838). Overall, 85% of participants who completed the 360° turn test in >10 steps were participants with PIGD and 73% of participants who completed the 360° turn test in >7 steps were participants with PIGD. Either cutoff value could be used depending on the clinical context. Cutoff values with similar sensitivity and specificity were identified for time to complete the 360° turn test. A cutoff of >1 interruption during tandem walk could also identify people with PIGD with good sensitivity (0.808), although accuracy was reduced compared to the 360° turn test: only 64% of participants with >1 interruption during tandem walk were PIGD.

Post hoc analyses

To investigate combinations of balance tests, we combined the steps-to-turn, the variable with the highest specificity, and the interruptions during the tandem walk, the variable with the highest sensitivity as per Criterion 1. If participants exceeded both Criterion 1 cutoffs of 10 steps to turn and 2 interruptions for tandem walk, the sensitivity was 0.46 and the specificity increased to 0.97.

Additional post hoc analyses demonstrated that the OC values from Criterion 2 (Table 2) were preferable for identifying PIGD phenotype in those with early PD. By the OCs in Criterion 2, only 6 participants with early PD labeled as PIGD by the MDS-UPDRS were misclassified as TD by the 360° Turn Test (number of steps and time to turn, collectively). By Criterion 1, 20 participants with early PD labeled as PIGD by the UPDRS were classified as TD by the 360° Turn Test (number of steps and time to turn, collectively; Table 3).

DISCUSSION

We investigated the value of functional tests for distinguishing between TD and PIGD phenotypes in persons with PD. The 360° turn test was best able to distinguish PIGD from TD subtype with high sensitivity, derived from Criterion 2, using both number of steps and time. Thus, the 360° turn test could serve as a quick and easily administered screening tool for clinicians to identify persons with PIGD. Although it may be better to misclassify persons with TD as PIGD, we cannot strongly recommend substituting the 360° turn test for the MDS-UPDRS test, because the specificity was low. Nevertheless, if time is short and the therapist does not have a score available for the MDS-UPDRS, and the patient will not see their neurologist for several months, the 360° turn test may provide valuable insight into prognosis and trajectory.

Because of the trade-off between specificity and sensitivity, it is recommended to assess people who are initially positive with an additional test with low sensitivity and high specificity.[34] For Criterion 2, the reactive postural control test (specificity=0.686) had the

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greatest specificity, which was still low. Therefore, further investigation is needed to identify another brief physical test that could differentiate PIGD from TD phenotype with higher specificity. To identify as many persons with PIGD as possible, we recommend using the more sensitive OCs of 7 steps and 3.67 s needed for the 360° turn test so a person who takes more steps or time than the cutoff value would be classified as PIGD.

Individuals with TD and PIGD phenotype performed similarly on reactive postural control and the tandem test, limiting the use of these tests to determine subtype. Furthermore, sensitivity in ROC curve analysis of reactive postural control was only 0.579. However, number of interruptions during tandem walk was the most sensitive test using Criterion 1. In fact, administering both the tandem walk and the 360° turn test leads to very high specificity (Specificity=0.97) in identifying those who are PIGD. Other studies have shown conflicting results. Tandem walking has been validated as a predictor of recurrent falls in one study whereas another demonstrated no correlation between tandem walk and tests assessing clinical balance and executive function, and one-leg stance was moderately able to distinguish PIGD from TD (specificity=0.676).[35, 36]

Turning is impaired in people with PD, with one study showing people with PD need seven steps to turn.[37] Compared to five steps normally required by older adults, increased steps needed to turn has been correlated to lower UPDRS and Self-assessed Disability Scale (SAS) scores.[38] Turning difficulties are associated with poor quality of life and cognitive functioning,[39] stability during turns is not improved by medication,[40] and turns are a sensitive predictor of two key symptoms of PD locomotion predominant in persons with PIGD: freezing[38] and falling.[41] The 360° turn test captures these turning deficits, which may explain why this test was better able to distinguish between PD subtypes than other variables tested.[20]

The motor tests examined included both dynamic and static balance, activities which activate distinct regions of the brain. The sensorimotor cortex in involved in normal gait, [42, 43] and the supplementary motor area and prefrontal cortex are associated with walking speed.[44] The 360° turn test may be able to differentiate between PD phenotypes because this test captures functional impacts of different patterns of neurodegeneration between phenotypes. At the same disease stages and duration, persons with PIGD exhibit more white matter degradation, supporting the idea of TD being a less pathological PD subtype.[45] Given that the 360° turn test provided the most sensitive discriminatory ability, the PIGD subtype, which likely results from a different pattern of neurodegeneration from TD, may have greater impact on dynamic than static balance activities. As such, post hoc analyses determined that Criterion 2, which is more heavily weighted for sensitivity, distinguishes subtypes better among individuals with early stage PD, with less neurodegeneration.

Differing therapeutic interventions are recommended for TD and PIGD phenotypes to address the phenotype-specific challenges associated with PD.[46] Clinicians and investigators with insufficient time to administer the more comprehensive and time consuming UPDRS or MDS-UPDRS, who are however interested in tailoring interventions to the individual would likely benefit from having shorter assessments to determine PD phenotype.

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Limitations

This study had several limitations. The 360° turn test was administered in real-time and an expert rater counted the steps and timed the turns with a stopwatch. Capturing this information with inertial sensors may lead to more accurate estimations of performance.[47] However, timing mobility measures, e.g., gait speed, can be accurately and reliably timed with a stopwatch.[48] Some participant-reported data were missing; participants either chose not to give the information or inadvertently skipped answering some items in questionnaires. Participants sometimes refused to perform some physical tests for various reasons, e.g.. they felt they could not attempt the tests. It is impossible to know if fear of falling or true balance impairment caused the refusal, however, balance confidence and functional mobility are independently associated with falls in people with PD.[49] We did not use imputations to not mislead interpretations.

Assessments were conducted in the “off” state, which may limit generalizability, as most clinical visits post-diagnosis are conducted during “on” time. Other balance measures not included in this study could be more specific or sensitive for differentiating between PIGD and TD phenotypes.

The internal validity may be compromised because our sample size was not big enough to split the data into training and testing sets. External validity is limited because of excluding indeterminate persons with PD. Furthermore, ROC analysis used to determine the OCs may limit the generalizability of results.[50, 51] No globally optimal combination of sensitivity and specificity for a classifier esists.[30, 31, 33, 51] To increase sensitivity in Criterion 2, we selected a weighting factor for sensitivity of 1.5 based on preliminary analyses that found this weighting to be reasonable, easy to remember and implement, and not prone to identification of degenerate solutions for OCs. More systematic investigation is required to better understand the influence of this parameter on PIGD identification.

Conclusion

The 360° turn test requires minimal time to administer and may be useful in mild-moderate PD for distinguishing PIGD from TD subtypes, particularly when comprehensive neurological examination is not feasible, will take considerable time to obtain, or when item-level motor exam results are unavailable. Conducting the 360° turn test may provide a simple but effective way to differentiate between subtypes as opposed to conducting the entire MDS-UPDRS/UPDRS assessment, allowing clinicians and persons with PD to conserve time while expediting the development of individual rehabilitation regimens.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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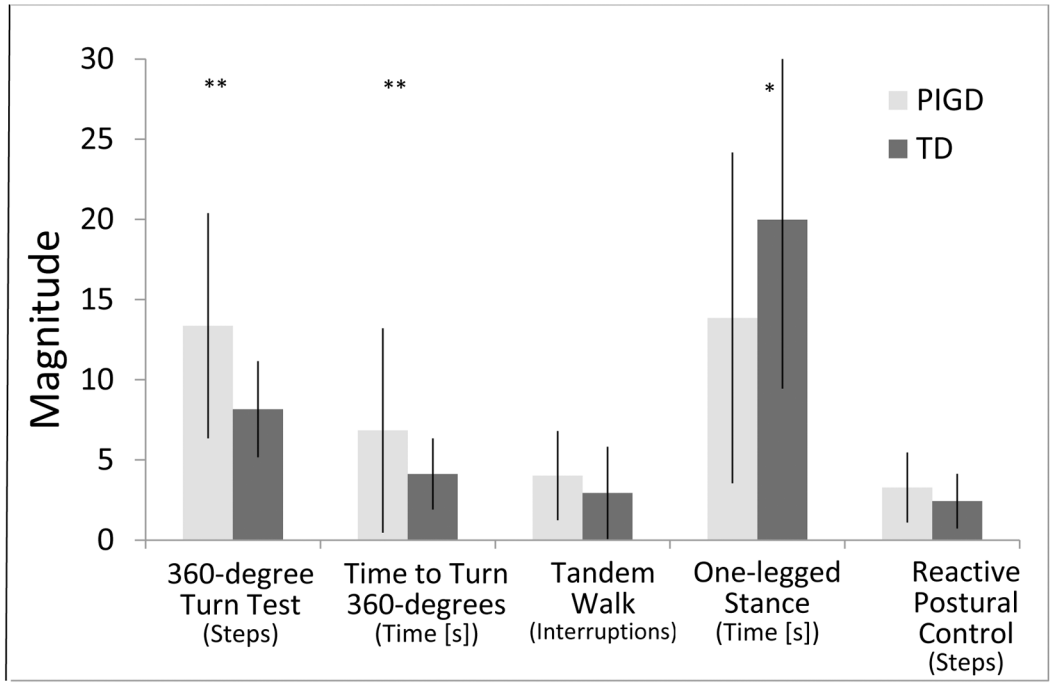


Figure 1. Tremor Dominant and Postural Insability/Gait Difficulty participants' performance on Physical Exams.

** p-value < 0.01, * p-value < 0.05. Values plotted are Means +/− Standard Deviation.

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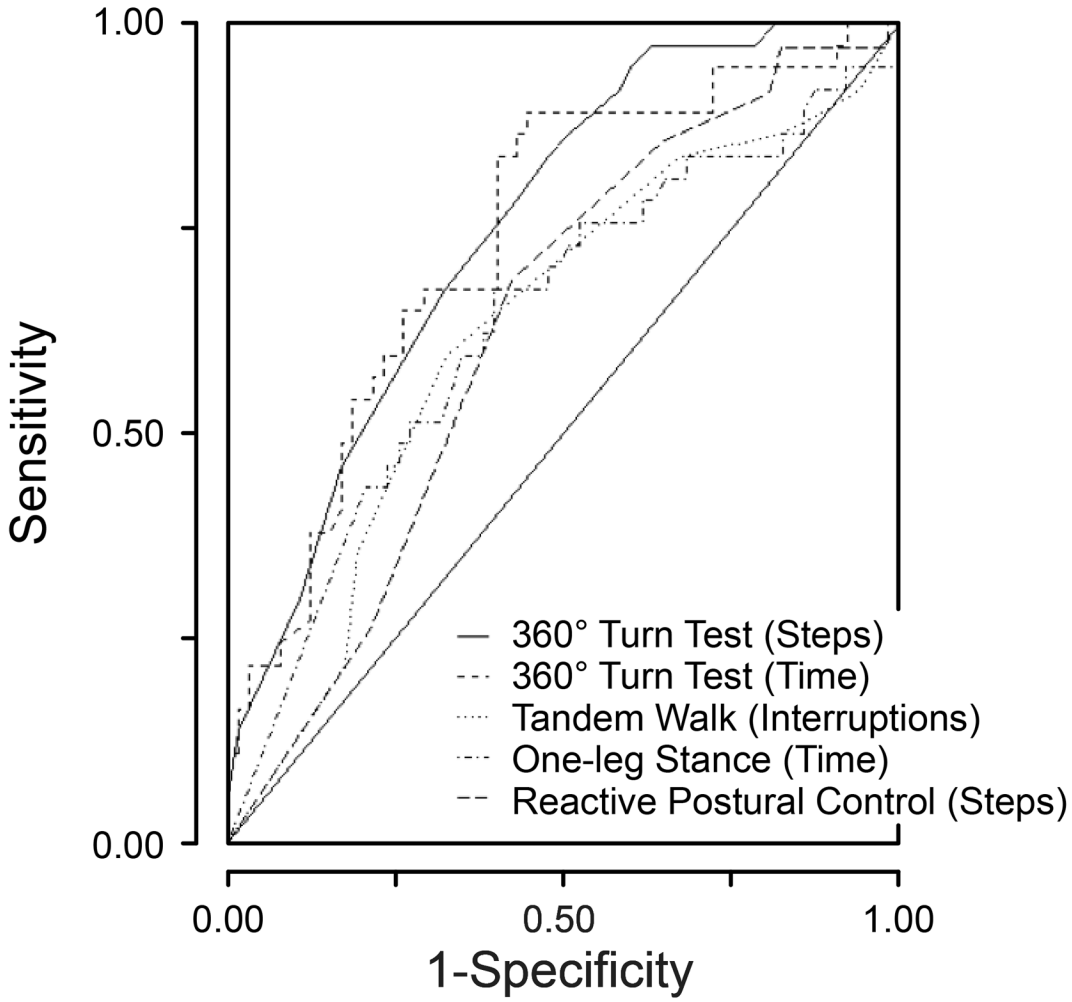


Figure 2.
Receiver Operating Characteristic (ROC) plots depicting the ability of balance variables to differentiate between PIGD and TD.

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Table 1.

Demographic Characteristics of Parkinson's Disease Motor Subgroups (Tremor Dominant and Postural Instability/Gait Difficulty).

Frequency of different characteristics may vary because of missing data. Missing data were excluded when calculating percentage, mean and SD. Fisher's exact tests were used for categorical variables, while two sample independent t test were used for continuous variables.

Characteristics	Whole Sample (n=104) Mean (SD)/ N(%)	PIGD (n=67) Mean (SD)/ N(%)	TD (n=37) Mean (SD)/ N(%)	P-value
Sex	n=104	n=67	n=37	
Female	43 (41.7%)	26 (38.8%)	17 (45.9%)	0.5357
Male	60 (58.3%)	41 (61.2%)	20 (54.1%)	
Age (years)	n=103	n=66	n=37	
	68.3 (8.8)	68.5 (8.2)	68.1 (9.8)	0.8330
Education (years)	n=102	n=65	n=37	
	13.1 (5.4)	12.5 (5.4)	14.1 (5.4)	0.1435
Number of Comorbidities	n=104	n=67	n=37	
	9.6 (10.6)	9.7 (10.7)	9.4 (10.5)	0.9023
Number of Medications	n=103	n=66	n=37	
	5.8 (3.8)	6.1 (3.7)	5.3 (3.9)	0.3735
Number of Years with PD	n=103	n=66	n=37	
	7.1 (5.3)	7.8 (5.6)	5.9 (4.6)	0.0631
Use of Assistive Device for Walking	n=102	n=65	n=37	
Yes	30 (29.4%)	25 (38.5%)	5 (13.5%)	0.0121 *
No	72 (70.6%)	40 (61.5%)	32 (86.5%)	
Have Fallen within the Past 6 Months	n=104	n=67	n=37	
Yes	60 (57.7%)	47 (70.1%)	13 (35.1%)	0.0008 ***
No	44 (42.3%)	20 (29.9%)	24 (64.9%)	
Number of Falls Past 6 Months [1]	n=104	n=67	n=37	
	9.07 (31.22)	13.3(38.8)	1.95 (5.0)	0.0207 *
Self-rated Quality of Life	n=102	n=65	n=37	
Low	6 (5.9%)	4 (6.2%)	2 (5.4%)	0.0194 *
Moderate	28 (27.5%)	24 (36.9%)	4 (10.8%)	
High	53 (52%)	30 (46.2%)	23 (62.2%)	
Very High	15 (14.7%)	7 (10.8%)	8 (21.6%)	
Freezing ≥ once/week	n= 99	n=62	n=37	
Yes	40 (40.4%)	32 (51.6%)	8 (21.6%)	0.0055 **
No	59 (59.6%)	30 (48.4%)	29 (78.4%)	
Hoehn & Yahr	n=104	n=67	n=37	
Stage 1	3 (2.9%)	1 (1.5%)	2 (5.4%)	0.0020 **
Stage 1.5	15 (14.4%)	7 (10.4%)	8 (21.6%)	

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Characteristics	Whole Sample (n=104) Mean (SD)/ N(%)	PIGD (n=67) Mean (SD)/ N(%)	TD (n=37) Mean (SD)/ N(%)	P-value
Stage 2	40 (38.5%)	21 (31.3%)	19 (51.4%)	
Stage 2.5	14 (13.5%)	9 (13.4%)	5 (13.5%)	
Stage 3	31 (29.8%)	28 (41.8%)	3 (8.1%)	
Stage 4	1 (1%)	1 (1.5%)	0 (0%)	
MDS-UPDRS				
Part I	n=74	n=46	n=28	
	13.5 (7.4)	15.6 (7.3)	10 (6.3)	0.0010 **
Part II	n=74	n=46	n=28	
	15.3 (7.9)	18 (7.4)	10.9 (6.9)	0.0001 ***
Part III	n=104	n=67	n=37	
	36.1 (11.8)	36.7 (12.2)	35 (11.2)	0.4834
Part IV	n=74	n=46	n=28	
	3.4 (3.6)	4 (3.8)	2.6 (3)	0.0964
Montreal Cognitive Assessment (MoCA) Score	n=104	n=67	n=37	
	25.6 (3.3)	24.9 (3.6)	26.8 (2.5)	0.0032 **

* p-value < 0.05

** p-value < 0.01

*** p-value < 0.001. Percentages are calculated within subtype groups.

[1] No Subject chose an option of ‘Less than once per week’

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Table 2:

Suggested Cutoff Values for Differentiating PD Phenotypes Using Motor Tests

	ROC Analysis			Criterion 1				Criterion 2			
Variable	AUC	CI	P-value	OC	Prob.†	Sen	Spe	OC	Prob.†	Sen	Spe
360° turn test (# of steps)	0.751	[0.658, 0.845]	0.001*	10	0.663	0.523	0.838	7	0.492	0.831	0.459
360° turn test (time (seconds))	0.740	[0.640, 0.841]	0.007*	5.01	0.648	0.554	0.892	3.67	0.538	0.815	0.541
Tandem walk test (# of interruptions)	0.625	[0.505, 0.745]	0.083	>2	0.551	0.673	0.595	>1	0.516	0.808	0.351
One-leg stance (s)	0.635	[0.520, 0.750]	0.017*	15.88	0.647	0.603	0.676	28.14	0.499	0.794	0.432
Reactive postural control (# of steps)	0.630	[0.517, 0.742]	0.061	2	0.601	0.579	0.686	_	_	_	_

Abbreviations: OC=Optimum Cutoff. Sen = sensitivity. Spe = specificity. Cutoff values are shown that maximize the sum of sensitivity and specificity (Criterion 1), and a weighted sum of sensitivity and specificity that weighted sensitivity 50% more heavily (Criterion 2). Cases in which both criteria resulted in the same cutoff values are indicated by "−. "

*Statistically significant at alpha= 0.05.

†Probability of being PIGD given the Optimal Cutoff is used as subtype classification criterion.

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Table 3:

Distribution of PIGD phenotype classification by the 360° turn test variables among study participants in Early PD (Hoehn &Yahr Stage 1, 1.5, and 2) (N=38).

	Variable: Number of Steps to Turn	Variable Time to Turn
Criterion 1 (N, %)		
Correctly classified participants	26 (68)	24 (63)
Misclassified	12 (32)	14 (36)
True TD misclassified as PIGD	4 (25)	2 (14)
True PIGD misclassified as TD	8 (75)	12 (86)
Criterion 2 (N, %)		
Correctly classified participants	22 (58)	25 (66)
Misclassified	16 (42)	12 (34)
True TD misclassified as PIGD	14 (88)	8 (75)
True PIGD misclassified as TD	2 (12)	4 (25)

Terms: “Correctly classified,” implies classification based on the balance variable was consistent with PIGD phenotype on MDS-UPDRS; “Misclassified,” implies classification based on the balance variable was inconsistent with PIGD phenotype on MDS-UPDRS.